

12.

(a.) Suppose that you wish to classify an observation $X \in \mathbb{R}$ into apples and oranges. You fit a logistic regression model and find that

$$\widehat{\Pr}(Y = \text{orange} \mid X = x) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)}.$$

Substitute the two probabilities:

$$\text{odds} = \frac{\frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)}}{\frac{1}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)}} = \exp(\hat{\beta}_0 + \hat{\beta}_1 x).$$

(b.) What is the log odds of orange versus apple in your friend's model?:

Using β notation, and given:

Using $\hat{\alpha}$ notation, and given:

$$\widehat{\Pr}(Y = \text{orange} \mid X = x) = \frac{\exp(\hat{\alpha}_{\text{orange}0} + \hat{\alpha}_{\text{orange}1}x)}{\exp(\hat{\alpha}_{\text{orange}0} + \hat{\alpha}_{\text{orange}1}x) + \exp(\hat{\alpha}_{\text{apple}0} + \hat{\alpha}_{\text{apple}1}x)}.$$

Step 1 : Calculate odds (orange vs. apple)

$$\text{odds} = \frac{\widehat{\Pr}(Y = \text{orange} \mid X = x)}{\widehat{\Pr}(Y = \text{apple} \mid X = x)} = \frac{\frac{\exp(\hat{\alpha}_{\text{orange}0} + \hat{\alpha}_{\text{orange}1}x)}{S}}{\frac{\exp(\hat{\alpha}_{\text{apple}0} + \hat{\alpha}_{\text{apple}1}x)}{S}} = \frac{\exp(\hat{\alpha}_{\text{orange}0} + \hat{\alpha}_{\text{orange}1}x)}{\exp(\hat{\alpha}_{\text{apple}0} + \hat{\alpha}_{\text{apple}1}x)},$$

where $S = \exp(\hat{\alpha}_{\text{orange}0} + \hat{\alpha}_{\text{orange}1}x) + \exp(\hat{\alpha}_{\text{apple}0} + \hat{\alpha}_{\text{apple}1}x)$ is elimited.

By the exponential calculation property, we can simplify the equation as following:

$$\text{odds} = \exp((\hat{\alpha}_{\text{orange}0} - \hat{\alpha}_{\text{apple}0}) + (\hat{\alpha}_{\text{orange}1} - \hat{\alpha}_{\text{apple}1})x).$$

Step 2 : take log (log-odds)

$$\log(\text{odds}) = (\hat{\alpha}_{\text{orange}0} - \hat{\alpha}_{\text{apple}0}) + (\hat{\alpha}_{\text{orange}1} - \hat{\alpha}_{\text{apple}1})x.$$

if we define the difference as following :

$$\hat{\beta}_0 \equiv \hat{\alpha}_{\text{orange}0} - \hat{\alpha}_{\text{apple}0}, \quad \hat{\beta}_1 \equiv \hat{\alpha}_{\text{orange}1} - \hat{\alpha}_{\text{apple}1},$$

we can get the same equation as mine in (a) :

$$\log\left(\frac{\Pr(Y = \text{orange} \mid X = x)}{\Pr(Y = \text{apple} \mid X = x)}\right) = \hat{\beta}_0 + \hat{\beta}_1x.$$

Although there are only 2 possible outcomes, we can get:

$$\widehat{\Pr}(Y = \text{apple} \mid X = x) = \frac{\exp(\hat{\alpha}_{\text{apple}0} + \hat{\alpha}_{\text{apple}1}x)}{\exp(\hat{\alpha}_{\text{orange}0} + \hat{\alpha}_{\text{orange}1}x) + \exp(\hat{\alpha}_{\text{apple}0} + \hat{\alpha}_{\text{apple}1}x)}.$$

(c) Suppose that in your model, $\hat{\beta}_0 = 2$ and $\hat{\beta}_1 = -1$. What are the coefficient estimates in your friend's model? Be as specific as possible.

Sol:

According the relationship we get between $\hat{\beta}$ and $\hat{\alpha}$ in(b) Step 2

$$\hat{\beta}_0 \equiv \hat{\alpha}_{\text{orange}0} - \hat{\alpha}_{\text{apple}0}, \quad \hat{\beta}_1 \equiv \hat{\alpha}_{\text{orange}1} - \hat{\alpha}_{\text{apple}1},$$

Equivalently,

$$(2, -1) = (\hat{\alpha}_{\text{orange}0} - \hat{\alpha}_{\text{apple}0}, \hat{\alpha}_{\text{orange}1} - \hat{\alpha}_{\text{apple}1}).$$

(d)Now suppose that you and your friend fit the same two models on a different data set. This time, your friend gets the coefficient estimates:

What are the coefficient estimates in your model?

Friend's coefficients: $(B_0, B_1) = (1.2 - 3, -2 - 0.6) = (-1.8, -2.6)$:

(e) Finally, suppose you apply both models from (d) to a data set with 2,000 test observations. What fraction of the time do you expect the predicted class labels from your model to agree with those from your friend's model? Explain your answer.

For binary classification, logistic regression and softmax models produce the same log-odds up to a constant shift, so predicted classes are identical. Therefore agreement=100%